WORKED SOLUTION

STEP I 2011, Question 6

Pure • Binomial Series Summation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q6

The key idea

Once we know the coefficients of $(1-x)^{-3}$, multiplying by a polynomial just mixes a few neighbouring coefficients together, so coefficients of more complicated rational functions come almost for free. Reading the process backwards, a series whose terms are (polynomial in r) $\times\, x^r$ can be summed by spotting which rational function generates it and then substituting the right value of x, here $x = \frac{1}{2}$.

Solution

The opening result

For $|x| < 1$ the binomial expansion with index -3 gives

$$(1-x)^{-3} = \sum_{r=0}^{\infty} \frac{(-3)(-4)\cdots(-3-r+1)}{r!}\,(-x)^r.$$

The numerator has r factors, each negative, so their product is $(-1)^r\,\dfrac{(r+2)!}{2!}$, and the $(-1)^r$ cancels with the one from $(-x)^r$. Hence the coefficient of x^r is

$$a_r = \frac{(r+2)!}{2!\,r!} = \tfrac{1}{2}(r+1)(r+2),$$

as required. (So $a_0 = 1$, $a_1 = 3$, $a_2 = 6$, $a_3 = 10$, …)

Part (i)

Write

$$\frac{1-x+2x^2}{(1-x)^3} = (1-x+2x^2)\sum_{k\geq 0} a_k x^k.$$

A term x^r arises from $1 \cdot a_r x^r$, from $-x \cdot a_{r-1}x^{r-1}$ and from $2x^2 \cdot a_{r-2}x^{r-2}$, so for $r \geq 2$ the coefficient of x^r is

$$\begin{aligned} a_r - a_{r-1} + 2a_{r-2} &= \tfrac{1}{2}(r+1)(r+2) - \tfrac{1}{2}r(r+1) + (r-1)r \\ &= (r+1) + r^2 - r \\ &= r^2 + 1. \end{aligned}$$

Ask yourself Does the formula still hold for $r = 0$ and $r = 1$, where a_{r-1} or a_{r-2} does not exist?

Yes. If we set $a_{-1} = a_{-2} = 0$, the formula $\frac{1}{2}(r+1)(r+2)$ also gives 0 at $r = -1$ and $r = -2$, so the calculation above is valid for every $r \geq 0$. Directly: the constant term is $a_0 = 1 = 0^2 + 1$, and the coefficient of x is $a_1 - a_0 = 2 = 1^2 + 1$. So the coefficient of x^r is $r^2 + 1$ for all $r \geq 0$, as required.

For the series, the numerators $1, 2, 5, 10, 17, 26, 37, 50, \ldots$ are $r^2 + 1$ for $r = 0, 1, 2, \ldots$ and the denominators are 2^r. So the series is

$$\sum_{r=0}^{\infty} (r^2 + 1) \left(\tfrac{1}{2}\right)^r,$$

which is the expansion above at $x = \frac{1}{2}$. This is legitimate because the binomial expansion of $(1-x)^{-3}$ converges for $|x| < 1$. Therefore

$$1 + \frac{2}{2} + \frac{5}{4} + \frac{10}{8} + \cdots = \frac{1 - \frac{1}{2} + 2 \cdot \frac{1}{4}}{\left(\frac{1}{2}\right)^3} = \frac{1}{1/8} = 8.$$

Part (ii)

The first job is to find the pattern. Rewriting the terms with denominators $1, 2, 4, 8, 16, 32, 64$:

$$1, \ \frac{4}{2}, \ \frac{9}{4}, \ \frac{16}{8}, \ \frac{25}{16}, \ \frac{36}{32}, \ \frac{49}{64}, \ \cdots$$

so the series is $\displaystyle\sum_{r=0}^{\infty} \frac{(r+1)^2}{2^r}$.

Ask yourself Which numerator over $(1-x)^3$ gives coefficients $(r+1)^2$?

Part (i) suggests trying a numerator $1 + x$ (a short polynomial again). The coefficient of x^r in $(1+x)(1-x)^{-3}$ is

$$a_r + a_{r-1} = \tfrac{1}{2}(r+1)(r+2) + \tfrac{1}{2}r(r+1) = \tfrac{1}{2}(r+1)(2r+2) = (r+1)^2,$$

valid for all $r \geq 0$ (with $a_{-1} = 0$ as before). Putting $x = \frac{1}{2}$,

$$1 + 2 + \frac{9}{4} + 2 + \frac{25}{16} + \cdots = \frac{1 + \frac{1}{2}}{\left(\frac{1}{2}\right)^3} = \frac{3}{2} \cdot 8 = 12.$$

Alternatively, using part (i). Shift the index with $s = r + 1$, so that $2^{-r} = 2 \cdot 2^{-s}$:

$$\sum_{r \geq 0} \frac{(r+1)^2}{2^r} = 2 \sum_{s \geq 1} \frac{s^2}{2^s} = 2 \left(\sum_{s \geq 0} \frac{s^2 + 1}{2^s} - \sum_{s \geq 0} \frac{1}{2^s} \right) = 2(8 - 2) = 12,$$

using the answer to part (i) and the geometric series $\sum 2^{-s} = 2$.

Checks

- Expanding $(1 - x)^{-3}$ by hand as $(1 + x + x^2 + \cdots)^3$, or by multiplying out a few terms, gives $1 + 3x + 6x^2 + 10x^3 + \cdots$, matching $\frac{1}{2}(r+1)(r+2)$.
- The partial sums of the part (i) series approach 8: after 200 terms the sum agrees with 8 to machine precision. The same holds for part (ii) with limit 12.
- The two routes in part (ii) give the same answer, 12.

Common slips

- Getting the sign wrong in the binomial coefficient: there are r minus signs from $(-3)(-4)\cdots$ and r from $(-x)^r$, and they cancel.
- In part (ii), misreading the pattern because some fractions are given in lowest terms: $2 = \frac{4}{2} = \frac{16}{8}$ and $\frac{9}{8} = \frac{36}{32}$.
- Substituting $x = \frac{1}{2}$ without noting that the expansion converges there ($|x| < 1$).

Beyond STEP

What we have been doing is using *generating functions*: a sequence $c_0, c_1, c_2, \ldots$ is encoded as the power series $\sum c_r x^r$, and operations on sequences become algebra on functions. In fact any polynomial in r times x^r sums to a rational function with denominator a power of $(1 - x)$, which is why these sums always come out rational. Justifying the substitution of a value of x rests on the theory of power series and their radius of convergence, covered in Abbott, *Understanding Analysis*; for generating functions as a problem-solving tool, see Siklos, *Advanced Problems in Mathematics*.

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